Volumes of Revolution

Section 1: Disk Method (Rotation Around Axes)

1 Find the volume of the solid generated by revolving $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

a. 8π

b. 16π

c. 4π

d. $32\pi/3$

2 Find the volume when the region under $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis.

a. $\frac{32\pi}{5}$

b. $\frac{16\pi}{5}$

c. 8π

d. $\frac{64\pi}{5}$

3 Find the volume when the region between $x = \sqrt{y}$, $y = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π

b. 16π

c. 4π

d. $32\pi/3$

Section 2: Washer Method (Hollow Solids)

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the volume when the region bounded by $y = x$ and $y = x^2$ is revolved about the x-axis.

a. $\frac{2\pi}{15}$

b. $\frac{\pi}{15}$

c. $\frac{4\pi}{15}$

d. $\frac{\pi}{5}$

5 The region bounded by $y = 4 - x^2$ and $y = 0$ is rotated about the line $y = -1$. Set up the outer and inner radii.

a. $R(x) = (4 - x^2) - (-1) = 5 - x^2$, $r(x) = 0 - (-1) = 1$

b. $R(x) = 4 - x^2$, $r(x) = 1$

c. $R(x) = 5 - x^2$, $r(x) = 0$

d. $R(x) = 4 - x^2 + 1$, $r(x) = 2$

6 Evaluate the volume in the previous problem rotated about $y = -1$ on $[-2, 2]$.

a. $\frac{384\pi}{15} = \frac{128\pi}{5}$

b. $\frac{64\pi}{5}$

c. 32π

d. $\frac{256\pi}{5}$

Section 3: Cylindrical Shells Method

7 State the cylindrical shells formula for a region under $y = f(x)$ on $[a, b]$ revolved about the y-axis.

a. $V = 2\pi \int_a^b x f(x)\, dx$

b. $V = \pi \int_a^b [f(x)]^2\, dx$

c. $V = 2\pi \int_a^b [f(x)]^2\, dx$

d. $V = \pi \int_a^b x^2 f(x)\, dx$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Find the volume when the region under $y = 3x - x^2$ from $x = 0$ to $x = 3$ is revolved about the y-axis.

a. $\frac{27\pi}{2}$

b. 27π

c. $\frac{81\pi}{4}$

d. $\frac{9\pi}{2}$

9 Use cylindrical shells to find the volume when $y = e^{-x^2}$ on $[0, 1]$ is revolved about the y-axis.

a. $\pi(1 - e^{-1})$

b. $2\pi(1 - e^{-1})$

c. πe^{-1}

d. $\frac{\pi}{2}(1 - e^{-1})$

Section 4: Cross-Sections with Known Geometric Shapes

10 The base of a solid is the circle $x^2 + y^2 = 4$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

a. $\frac{128}{3}$

b. 16π

c. 32

d. $\frac{64}{3}$

11 The base of a solid is bounded by $y = x^2$ and $y = 4$. Cross-sections perpendicular to the y-axis are semicircles. Find the volume.

a. 4π

b. 2π

c. 8π

d. $\frac{8\pi}{3}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 The base is the triangle with vertices $(0,0), (1,0), (0,1)$. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume.

a. $\frac{\sqrt{3}}{12}$

b. $\frac{\sqrt{3}}{6}$

c. $\frac{1}{6}$

d. $\frac{\sqrt{3}}{4}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647